

Propositional Logic Series

Episode 3: Worksheet

What Makes an Argument Valid?

Instructions

For each exercise, justify your answer. When an argument is invalid, give a counterexample: a case in which *all* premises are true and the conclusion is false. Do not decide validity merely by checking whether the displayed statements happen to be true.

Exercises

1. The counterexample criterion.

For each proposed case, decide whether it is sufficient to show that an argument is invalid. Briefly explain.

- (a) Every premise is true and the conclusion is false.
- (b) Some premises are true and the conclusion is false.
- (c) Every premise is true and the conclusion is true.
- (d) One premise is false and the conclusion is false.

2. Reading possible cases.

An argument has premises P_1, P_2 and conclusion C . Determine which rows, if any, are counterexamples.

Case	P_1	P_2	C
A	true	true	true
B	true	false	false
C	true	true	false
D	false	true	false

What can be concluded about the argument from the table?

3. A fixed mathematical argument.

If 18 is divisible by 6, then 18 is even.

18 is divisible by 6.

$\therefore$ 18 is even.

- (a) Is the argument valid or invalid?
- (b) Explain your answer using the counterexample criterion.
- (c) Is the argument sound?

4. A mathematical counterexample.

5 is prime or 9 is prime.
5 is prime.
 $\therefore$ 9 is prime.

Determine whether the argument is valid. If it is invalid, identify explicitly which premises are true and which conclusion is false.

5. True statements are not enough.

Suppose a card currently shows the number 12.

The number on the card is divisible by 3.
The number on the card is even.
 $\therefore$ The number on the card is greater than 10.

- (a) Are all three displayed statements true in the original case?
- (b) Does that establish validity?
- (c) Test the case in which the card shows 6.

6. Valid form with false content.

If 14 is a multiple of 7, then 14 is odd.
14 is a multiple of 7.
 $\therefore$ 14 is odd.

- (a) Which displayed statements are actually true and which are false?
- (b) Is the argument valid or invalid?
- (c) Is the argument sound?

7. Kinds of validity.

Classify the principal kind of validity illustrated by each argument as *nomological*, *conceptual*, or *formal*. Give one sentence of justification.

- (a) An information signal travels from point A to point B in one second. Therefore, the distance between A and B is no more than approximately 300,000 kilometres.
- (b) Mina is an ophthalmologist. Therefore, Mina is a doctor.
- (c) Either Leyla is a musician or Leyla is an architect. Leyla is not an architect. Therefore, Leyla is a musician.

8. Bivalence.

Under the classical bivalence assumption used in this episode, decide whether each description is admissible as a case.

- (a) P is true and Q is false.
- (b) P is both true and false.

- (c) P is neither true nor false.
- (d) P is false and Q is false.

9. Soundness.

Classify each argument as *sound* or *not sound*. Use the information given; do not assume more.

- (a) The argument is valid, and all its premises are actually true.
- (b) The argument is valid, but at least one premise is false.
- (c) Every displayed statement is actually true, but the argument is invalid.
- (d) The argument is invalid, and its conclusion is false.

For each not-sound classification, state which defining condition fails.

10. Deductive does not mean valid.

If the device is fully charged, then the screen turns on.
 The screen turns on.
 $\therefore$ The device is fully charged.

- (a) The argument is presented as deductive. Does that make it valid?
- (b) Construct a counterexample in which the screen is powered externally.
- (c) State the four-step counterexample-search method used in the episode.

Sources and Note

The conceptual background for this worksheet is informed by P. D. Magnus, Tim Button, Robert Trueman, and Richard Zach, *forall x: Calgary: An Introduction to Formal Logic*, with contributions by J. Robert Loftis and Aaron Thomas-Bolduc, Chapter 2.

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